

11 (long)

- (a) State Gauss' law for electrostatics, and derive an expression for the capacitance of an isolated conductive sphere of diameter d in air. [5]
- (b) Derive an expression for the electric field surrounding an infinitely long isolated straight wire of radius R in air with a static charge density of $\rho \text{ C m}^{-1}$. [5]
- (c) Explain the principle of linear superposition for potential fields. Derive an expression for the capacitance per unit length between two parallel straight wires, each of radius R with centres separated by distance d (where $d \gg R$). [10]
- (d) Two wires 20 mm in diameter with centre-to-centre separation 2 m in air are to be used to carry high voltage. What is the capacitance per unit length between the wires? If the breakdown field of air is $> 10^6 \text{ V m}^{-1}$, what is the maximum voltage the wires can carry? [10]